# Project 1 - Explainable AI in Radiology

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## Abstract

This is a great paper, and it has a concise abstract.

**Keywords:** Chest X-Ray, Pleural Effusion, Explainable AI, Saliency maps, GradCAM, Example-base explanations, Cosine Similarity

## 1. Introduction

Machine learning has become a powerful tool for solving complex problems in various domains and is increasingly integrated into different applications. However, many of these models act as black boxes with limited knowledge of how they work and what led to a certain output(Ahmed et al., 2022) . Explainable AI (XAI) aims to improve the understanding of these systems by making the decision process more transparent. Healthcare, specifically radiology imaging, is one area with increased usage of AI systems to support diagnostic and treatment decisions(E. Ihongbe et al., 2024) . To ensure trust in these systems, especially when crucial medical decisions must be made with their help, it becomes increasingly important to not only produce efficient and accurate models but also explainable ones to increase trust in the model's outputs. Many AI tools in medicine are meant to support medical personnel in their decision processes and decrease their workload. Explainable models will increase the personnel's confidence in the decisions made by the system and let them detect any potential errors. One of the current main applications for AI in medicine is the analysis of X-ray images to detect various illnesses, most often utilized for bone or lung diseases (Miró-Nicolau et al., 2022) . Chest Radiography (CXR) images are one example of this technique. With many illnesses affecting the chest area and lungs, accurate systems are vital. They have become more relevant since the COVID-19 outbreak(Miró-Nicolau et al., 2022) , and people have not only realized how important good systems are but also the importance of being able to understand the system's decision through explainability. In this paper we will present the application of machine learning to X-ray images for predicting pleural effusion (a buildup of excess fluid in the lungs) and two explanation methods: Saliency maps and Cosine Similarity. Additionally, these methods are evaluated to verify their performance using Quantus(Hedström et al., 2023) and (—–).

## 2. Methods & Materials

### 2.1. Model

### 2.2. Saliency Maps

### 2.3. Example-based Explanations

## 3. Results & Discussion

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### 3.1. Saliency Maps

### 3.2. Example-based Explanations

## 4. Conclusion